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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

CV2302-1 – DESIGN OF BRIDGES
(For CV)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: i) Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

ii) Use of code of practice IRC 6, IRC 21, IS 458, IS 456 and Pigeaud's chart is permitted

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which of these is a movable bridge?
A) Tied arch
B) Cantilever bridge
C) Bascule
D) Beam bridge
2. The wing wall is part of:
A) Foundation
B) Drainage system
C) Superstructure
D) Substructure
3. Cofferdam is primarily used for:
A) Supporting bridge bearings
B) Temporary dewatering for foundations
C) Allowing expansion joint
D) Providing aesthetic finish
4. Which foundation type is most used where soil is weak and deep bearing strata exist?
A) Shallow footing
B) Raft foundation
C) Pile foundation
D) Combined Footing
5. Which of the following is considered in IRC for lane loading?
A) Only a single point load
B) Truck wheel loads and lane load
C) Earthquake effects only
D) Pedestrian live load only
6. When designing bridges as per IRC, the design vehicle for rural highways is generally:
A) Class AA only
B) Railway load
C) Class B
D) Class A or Class 70R/100R
7. The impact factor increases when:
A) Span length decreases
B) Span length increases
C) Vehicle speed decreases
D) Temperature is high
8. Which code gives specifications for plain and reinforced concrete used in bridge components in India?
A) IS 800
B) BS 8110
C) IS 456
D) AASHTO
9. When computing the required waterway opening for culverts, designer must consider:
A) Design discharge and permissible afflux
B) Ambient temperature and humidity
C) Paint type
D) Expansion joints
10. The design headwater for a culvert commonly refers to:
A) Minimum water level downstream of the culvert
B) Maximum water level upstream of the culvert
C) Flow velocity through the culvert
D) Depth of culvert barrel
11. For pipe culverts under embankment, bedding and backfill are important to:
A) Reduce ambient temperature around the pipe
B) Provide uniform support and prevent pipe deformation
C) Control water pH in the culvert
D) Increase flow velocity through the pipe

12. IS 458 relates to:
 A) Design of steel bridges B) Portland Pozzolana cement
 C) Cement concrete and reinforced concrete for exposure conditions D) Highway Geometric Design
13. For designing interior slab panels using Pigeaud's method, the slab panel is idealized into:
 A) Triangular elements for finite element analysis B) Rectangular plates with boundary conditions for moments
 C) Single cantilever spanning in one direction D) Circular panels with radial strips
14. In T-beam design, negative moment regions require:
 A) No reinforcement B) Bottom steel reinforcement
 C) Only shear reinforcement D) Top steel reinforcement
15. Courbon's method uses which assumption about load distribution?
 A) Loads spread uniformly to all girders regardless of spacing B) Load is triangularly distributed from support to midspan
 C) Load is ignored D) Only dead load is considered
16. For interior slab panels, aspect ratio (long side/short side) affects:
 A) Only deck finishing B) Soil bearing
 C) Moment distribution between short and long directions D) Bearing selection
17. Expansion joints are provided to:
 A) Reduce vehicle speed on the bridge B) Allow thermal expansion and contraction
 C) Reduce scour D) Replace bearings
18. For skew bridges, bearing design must consider:
 A) Only live load, ignoring dead load B) Temperature variations only
 C) Only vertical loads D) Eccentricity of load and combined rotation/translation effects
19. A rocker bearing primarily provides:
 A) Elastic deformation B) Hydraulic damping
 C) Complete fixity in all directions D) A pivoting action to facilitate rotation
20. Elastomeric bearings for girder bridges are often reinforced with:
 A) Steel plates B) Aluminum sheets
 C) Plastic beads D) Ceramic tiles

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Draw a neat sketch of a typical bridge and name the components.	8	L*2	1	1
b)	Explain the concept of impact factor in bridges and describe the procedure for its determination as per IRC standards.	8	L2	2	1
2. a)	Write an explanatory note on linear water way, afflux and their significance in design of bridges.	8	L2	1	1
b)	List various types of foundation, explain any one in detail.	8	L2	2	1
3. a)	Write the factors to be considered in selection of site for a bridge.	8	L1	1	1
b)	List the loads, forces, and stresses to be considered in the bridge design.	8	L1	2	1
Unit – II					
4. a)	Write a short note on Pigeaud's curves, discuss their utilities and limitations.	8	L3	4	1,2
b)	Obtain the expression for Courbon's reaction factor and discuss its limitations.	8	L3	4	1,2,3
5. a)	Briefly explain the steps involved in the design of a pipe culvert.	8	L3	3	1,2,3

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- b) Obtain the Courbon's reaction factor and maximum bending moment due to live load in case of a T-beam bridge having the following details:
 Road way 7.5m with 1m footpath on either side
 Live load IRC class AA tracked vehicle
 Number of girders are 4 with c/c spacing of 2.5m
 Span of the bridge = 25m
- 8 L3 4 1,2,3
6. Design the pipe culvert using the following data:
 Discharge in stream = $5\text{m}^3/\text{s}$; Road embankment height = 6m
 Road width = 7.5m and formation width is 10m
 Safe velocity is 3m/s , Density of soil = 20kN/m^3 , Side slope 1.5:1
 Loading = IRC class AA tracked vehicle
 Assume bell mouthed entry, $C_e = 1.5$, $C_s = 0.010$
- 16 L3 3 1,2,3
- Unit – III**
7. Which are the different types of joints used in bridges? Also, explain the functions of expansion joints.
- 16 L2 5 1
8. Explain any three types of bearings with neat sketches.
- 16 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
